

# Naïve Bayes Classifier

Adapted from slides from U. of  
Edinburgh and online sources

# Outline

- Bayes rule
- Discriminative vs. Generative Classifier
- Naïve Bayes classifier
- Examples
  - Discrete values
  - Continuous values: assuming normal distribution
- Issues
  - Zero values: smoothing
  - Missing values: fill the average

# Bayes Formula



$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

**Bayes, Thomas (1763)** An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, 53:370-418

Discriminative model

Posterior

Likelihood

Generative model

Prior

$$P(Y|X_1, \dots, X_n) = \frac{P(X_1, \dots, X_n|Y)P(Y)}{P(X_1, \dots, X_n)}$$

Normalization Constant

# Bayes Example

The entire output of a factory is produced on three machines. The three machines account for 20%, 30%, and 50% of the factory output. The fraction of defective items produced is 5% for the first machine; 3% for the second machine; and 1% for the third machine. If an item is chosen at random from the total output and is found to be defective, what is the probability that it was produced by the third machine?

# Solution

$$P(X_A) = 0.2, \quad P(X_B) = 0.3, \quad P(X_C) = 0.5.$$

If the item was made by the first machine, then the probability that it is defective is 0.05; that is,  $P(Y | X_A) = 0.05$ .

$$P(Y | X_A) = 0.05, \quad P(Y | X_B) = 0.03, \quad P(Y | X_C) = 0.01.$$

To answer the original question, we first find  $P(Y)$ . That can be done in the following way:

$$P(Y) = \sum_i P(Y | X_i)P(X_i) = (0.05)(0.2) + (0.03)(0.3) + (0.01)(0.5) = 0.024.$$

Hence 2.4% of the total output of the factory is defective.

We are given that  $Y$  has occurred, and we want to calculate the conditional probability of  $X_C$ . By Bayes' theorem,

$$P(X_C | Y) = \frac{P(Y | X_C)P(X_C)}{P(Y)} = \frac{0.01 \cdot 0.50}{0.024} = \frac{5}{24}$$

# Generative vs. Discriminative Classifier

Training classifiers involves estimating  $f: X \rightarrow Y$ , or  $P(Y|X)$

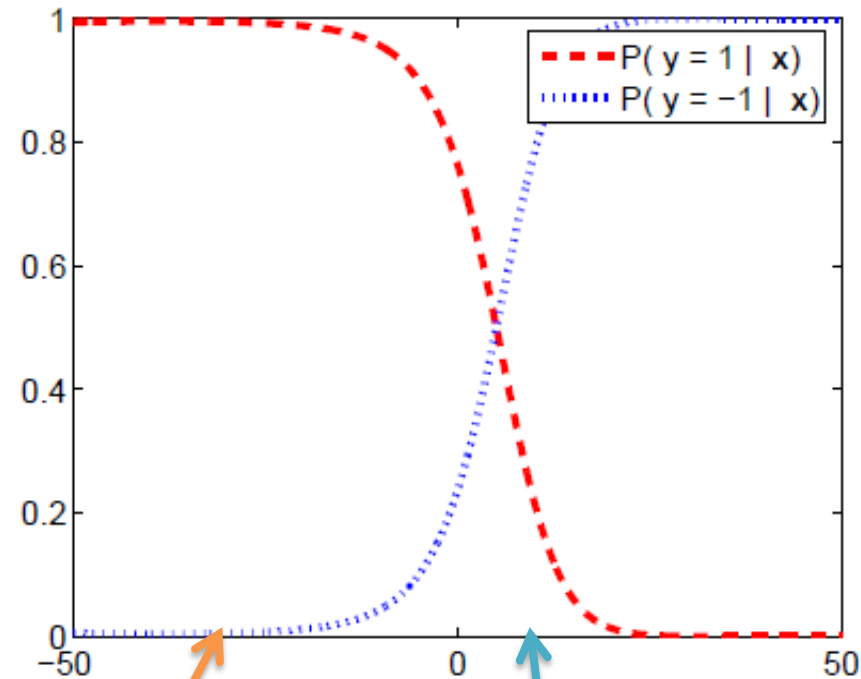
- Discriminative classifiers :
  - Assume some functional form for  $P(Y|X)$
  - Estimate parameters of  $P(Y|X)$  directly from training data
- Generative classifiers
  - Assume some functional form for  $P(X|Y)$ ,  $P(X)$
  - Estimate parameters of  $P(X|Y)$ ,  $P(X)$  directly from training data
  - Use Bayes rule to calculate  $P(Y|X = x_i)$

# Discriminative Model

- Logistic Regression

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + \exp(yf(\mathbf{x}))}$$

$$f^*(\mathbf{x}) = \begin{cases} +\infty & \Pr(y = 1|\mathbf{x}) > \frac{1}{2}, \\ -\infty & \Pr(y = -1|\mathbf{x}) < \frac{1}{2}, \\ \text{arbitrary} & \text{otherwise.} \end{cases}$$



- Color
- Size
- Texture
- Weight
- ...

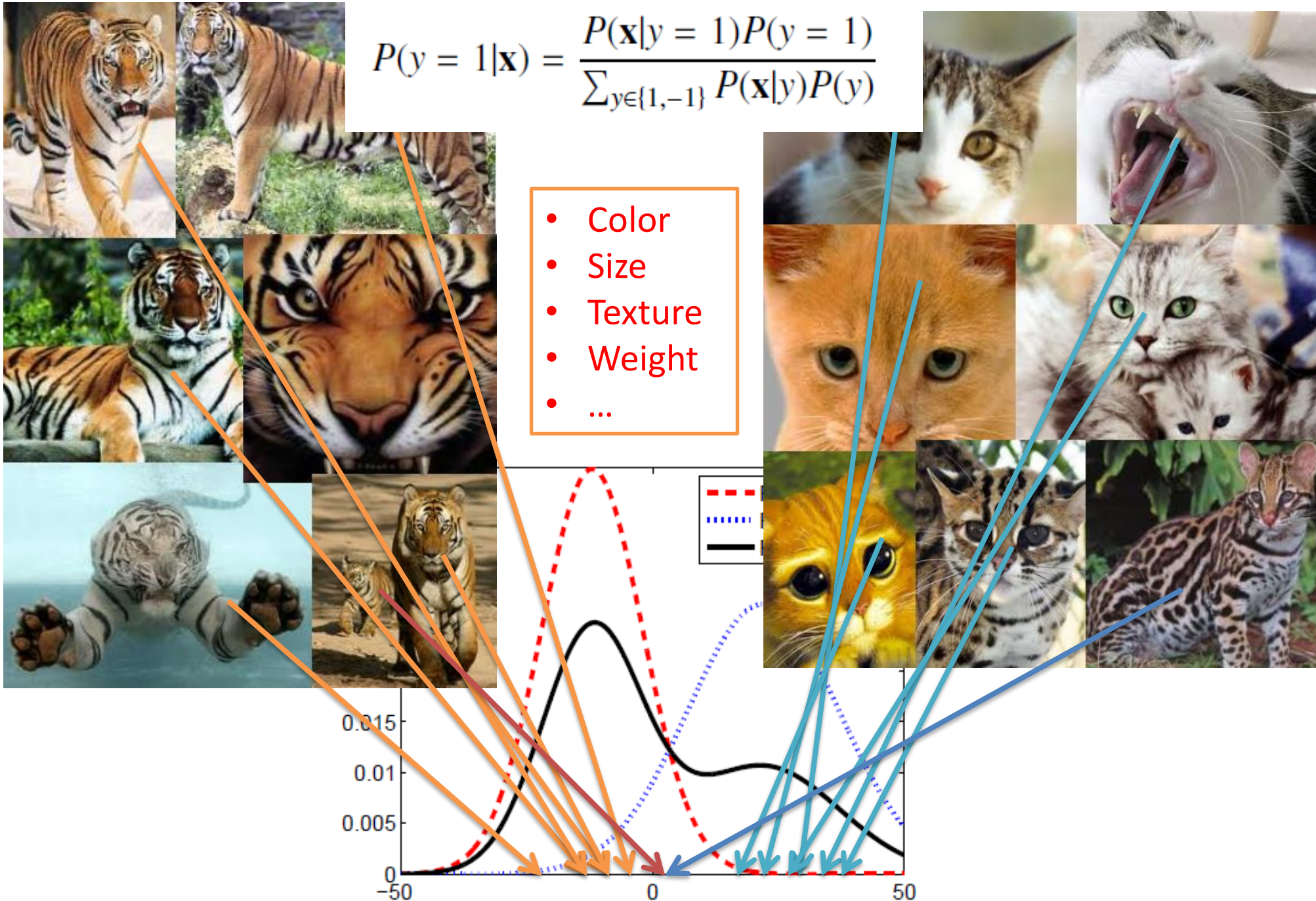




# Generative Model

$$P(y = 1|\mathbf{x}) = \frac{P(\mathbf{x}|y = 1)P(y = 1)}{\sum_{y \in \{1, -1\}} P(\mathbf{x}|y)P(y)}$$

- Color
- Size
- Texture
- Weight
- ...

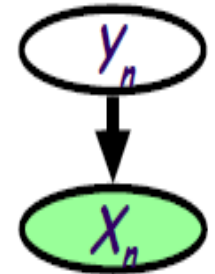




# Comparison

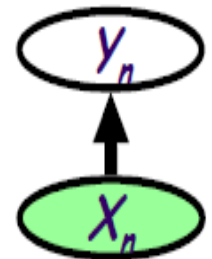
- Generative models

- Assume some functional form for  $P(X|Y)$ ,  $P(Y)$
- Estimate parameters of  $P(X|Y)$ ,  $P(Y)$  directly from training data
- Use Bayes rule to calculate  $P(Y|X=x)$



- Discriminative models

- Directly assume some functional form for  $P(Y|X)$
- Estimate parameters of  $P(Y|X)$  directly from training data



# Probability Basics

- Prior, conditional and joint probability for random variables
  - Prior probability:  $P(X)$
  - Conditional probability:  $P(X_1 | X_2), P(X_2 | X_1)$
  - Joint probability:  $\mathbf{X} = (X_1, X_2), P(\mathbf{X}) = P(X_1, X_2)$
  - Relationship:  $P(X_1, X_2) = P(X_2 | X_1)P(X_1) = P(X_1 | X_2)P(X_2)$
  - Independence:  $P(X_2 | X_1) = P(X_2), P(X_1 | X_2) = P(X_1), P(X_1, X_2) = P(X_1)P(X_2)$
- Bayesian Rule

$$P(C | \mathbf{X}) = \frac{P(\mathbf{X} | C)P(C)}{P(\mathbf{X})}$$

$$Posterior = \frac{Likelihood \times Prior}{Evidence}$$

# Probability Basics

- **Quiz:** We have two six-sided dice. When they are tolled, it could end up with the following occurrence: (*A*) dice 1 lands on side "3", (*B*) dice 2 lands on side "1", and (*C*) Two dice sum to eight. Answer the following questions:

1)  $P(A) = ?$

2)  $P(B) = ?$

3)  $P(C) = ?$

4)  $P(A | B) = ?$

5)  $P(C | A) = ?$

6)  $P(A, B) = ?$

7)  $P(A, C) = ?$

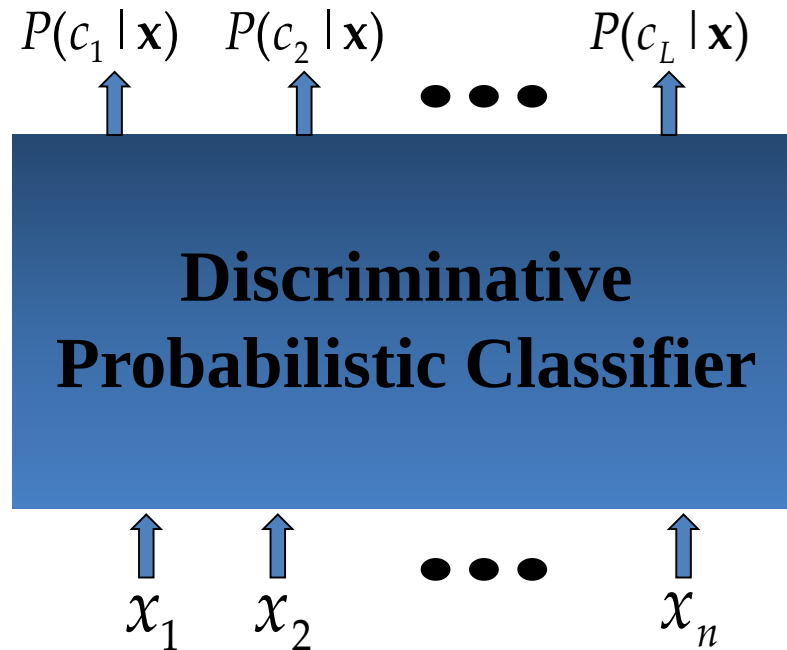
8) Is  $P(A, C)$  equals  $P(A) * P(C)$ ?



# Probabilistic Classification

- Establishing a probabilistic model for classification
  - Discriminative model**

$$P(C | \mathbf{X}) \quad C = c_1, \dots, c_L, \mathbf{X} = (X_1, \dots, X_n)$$

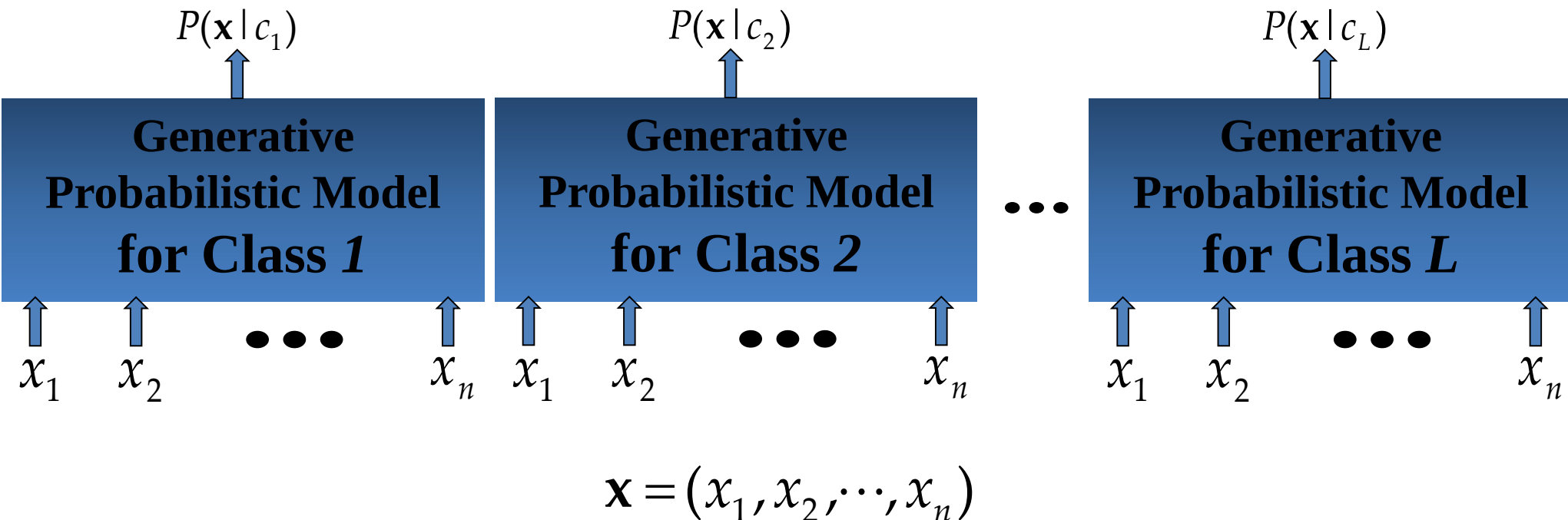


$$\mathbf{x} = (x_1, x_2, \dots, x_n)$$

# Probabilistic Classification

- Establishing a probabilistic model for classification (cont.)
  - Generative model**

$$P(\mathbf{X}|C) \quad C = c_1, \dots, c_L, \mathbf{X} = (X_1, \dots, X_n)$$



# Bayesian Classification

- Goal: learning function  $f(x) \rightarrow y$ 
  - $y$  ... one of  $k$  classes (e.g. spam/ham, digit 0-9)
  - $x = x_1 \dots x_n$  – values of attributes (numeric or categorical)
- Probabilistic classification:
  - most probable class given observation:  $\hat{y} = \arg \max_y P(y|x)$
- Bayesian probability of a class:

$$P(y|x) = \frac{\overbrace{P(x|y)}^{\text{class model}} \overbrace{P(y)}^{\text{prior}}}{\underbrace{\sum_{y'} P(x|y') P(y')}_{\text{evidence}}}$$



# Bayesian classification: components

$$P(y|x) = \frac{P(x|y)P(y)}{\sum_{y'} P(x|y')P(y')}$$

Example:

y ... patient has Avian flu

x ... observed symptoms

- $P(y)$ : prior probability of each class
  - encodes how which classes are common, which are rare
    - apriori much more likely to have common cold than Avian flu
- $P(x|y)$ : class-conditional model
  - describes how likely to see observation x for class y
  - assuming it's Avian flu, do the symptoms look plausible?
- $P(x)$ : normalize probabilities across observations

# The Naïve Bayes Model

- The *Naïve Bayes Assumption*: Assume that all features are independent **given the class label Y**
- Equationally speaking:

$$P(X_1, \dots, X_n | Y) = \prod_{i=1}^n P(X_i | Y)$$

- (We will discuss the validity of this assumption later)

# Naïve Bayes Training

- Training in Naïve Bayes is **easy**:
  - Estimate  $P(Y=v)$  as the fraction of records with  $Y=v$

$$P(Y = v) = \frac{\text{Count}(Y = v)}{\# \text{ records}}$$

- Estimate  $P(X_i=u|Y=v)$  as the fraction of records with  $Y=v$  for which  $X_i=u$

$$P(X_i = u|Y = v) = \frac{\text{Count}(X_i = u \wedge Y = v)}{\text{Count}(Y = v)}$$

- (This corresponds to Maximum Likelihood estimation of model parameters)

# Naïve Bayes Training

- In practice, some of these counts can be zero
- Fix this by adding “virtual” counts:

$$P(X_i = u|Y = v) = \frac{\text{Count}(X_i = u \wedge Y = v) + 1}{\text{Count}(Y = v) + 2}$$

- (This is like putting a prior on parameters and doing MAP estimation instead of MLE)
- This is called *Smoothing*

# Example

- Example: Play Tennis

*PlayTennis: training examples*

| Day | Outlook  | Temperature | Humidity | Wind   | PlayTennis |
|-----|----------|-------------|----------|--------|------------|
| D1  | Sunny    | Hot         | High     | Weak   | No         |
| D2  | Sunny    | Hot         | High     | Strong | No         |
| D3  | Overcast | Hot         | High     | Weak   | Yes        |
| D4  | Rain     | Mild        | High     | Weak   | Yes        |
| D5  | Rain     | Cool        | Normal   | Weak   | Yes        |
| D6  | Rain     | Cool        | Normal   | Strong | No         |
| D7  | Overcast | Cool        | Normal   | Strong | Yes        |
| D8  | Sunny    | Mild        | High     | Weak   | No         |
| D9  | Sunny    | Cool        | Normal   | Weak   | Yes        |
| D10 | Rain     | Mild        | Normal   | Weak   | Yes        |
| D11 | Sunny    | Mild        | Normal   | Strong | Yes        |
| D12 | Overcast | Mild        | High     | Strong | Yes        |
| D13 | Overcast | Hot         | Normal   | Weak   | Yes        |
| D14 | Rain     | Mild        | High     | Strong | No         |

# Example

- Training Phase

| Outlook         | Play=Yes | Play=No |
|-----------------|----------|---------|
| <i>Sunny</i>    | 2/9      | 3/5     |
| <i>Overcast</i> | 4/9      | 0/5     |
| <i>Rain</i>     | 3/9      | 2/5     |

| Temperature | Play=Yes | Play=No |
|-------------|----------|---------|
| <i>Hot</i>  | 2/9      | 2/5     |
| <i>Mild</i> | 4/9      | 2/5     |
| <i>Cool</i> | 3/9      | 1/5     |

| Humidity      | Play=Yes | Play=No |
|---------------|----------|---------|
| <i>High</i>   | 3/9      | 4/5     |
| <i>Normal</i> | 6/9      | 1/5     |

| Wind          | Play=Yes | Play=No |
|---------------|----------|---------|
| <i>Strong</i> | 3/9      | 3/5     |
| <i>Weak</i>   | 6/9      | 2/5     |

$$P(\text{Play=Yes}) = 9/14 \quad P(\text{Play=No}) = 5/14$$

# Example

- Testing Phase

- Given a new instance,

$\mathbf{x}' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

- Look up tables

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{Yes}) = 2/9$$

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{No}) = 1/5$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{No}) = 4/5$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Play}=\text{No}) = 5/14$$

- MAP rule

$$P(\text{Yes} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{Yes})P(\text{Cool} \mid \text{Yes})P(\text{High} \mid \text{Yes})P(\text{Strong} \mid \text{Yes})]P(\text{Play}=\text{Yes}) = 0.0053$$

$$P(\text{No} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{No})P(\text{Cool} \mid \text{No})P(\text{High} \mid \text{No})P(\text{Strong} \mid \text{No})]P(\text{Play}=\text{No}) = 0.0206$$

Given the fact  $P(\text{Yes} \mid \mathbf{x}') < P(\text{No} \mid \mathbf{x}')$ , we label  $\mathbf{x}'$  to be “No”.

# Relevant Issues

- Violation of Independence Assumption
  - For many real world tasks,  $P(X_1, \dots, X_n | C) \neq P(X_1 | C) \dots P(X_n | C)$
  - Nevertheless, naïve Bayes works surprisingly well anyway!
- Zero conditional probability Problem
  - If no example contains the attribute value  $X_j = a_{jk}$ ,  $\hat{P}(X_j = a_{jk} | C = c_i) = 0$
  - In this circumstance,  $\hat{P}(x_1 | c_i) \dots \hat{P}(a_{jk} | c_i) \dots \hat{P}(x_n | c_i) = 0$  during test
  - For a remedy, conditional probabilities estimated with

$$\hat{P}(X_j = a_{jk} | C = c_i) = \frac{n_c + mp}{n + m}$$

$n_c$  : number of training examples for which  $X_j = a_{jk}$  and  $C = c_i$

$n$  : number of training examples for which  $C = c_i$

$p$  : prior estimate (usually,  $p = 1/t$  for  $t$  possible values of  $X_j$ )

$m$  : weight to prior (number of "virtual" examples,  $m \geq 1$ )



# Naïve Bayes Smoothing

- In practice, some of these counts can be zero
- Fix this by adding “virtual” counts:

$$P(X_i = u|Y = v) = \frac{\text{Count}(X_i = u \wedge Y = v) + 1}{\text{Count}(Y = v) + 2}$$

- (This is like putting a prior on parameters and doing MAP estimation instead of MLE)
- This is called *Smoothing*

# Relevant Issues

- Continuous-valued Input Attributes
  - Numberless values for an attribute
  - Conditional probability modeled with the normal distribution

$$\hat{P}(X_j | C = c_i) = \frac{1}{\sqrt{2\pi}\sigma_{ji}} \exp\left(-\frac{(X_j - \mu_{ji})^2}{2\sigma_{ji}^2}\right)$$

$\mu_{ji}$  : mean (average) of attribute values  $X_j$  of examples for which  $C = c_i$

$\sigma_{ji}$  : standard deviation of attribute values  $X_j$  of examples for which  $C = c_i$

- Learning Phase: for  $\mathbf{X} = (X_1, \dots, X_n)$ ,  $C = c_1, \dots, c_L$   
Output:  $n \times L$  normal distributions and  $P(C = c_i)$   $i = 1, \dots, L$
- Test Phase: for  $\mathbf{X}' = (X'_1, \dots, X'_n)$ 
  - Calculate conditional probabilities with all the normal distributions
  - Apply the MAP rule to make a decision

# Common Practice for Continuous Value Problems

- The Naive Bayes Classifier for Data Sets with Numerical Attribute Values
- One common practice to handle numerical attribute values is to assume normal distributions for numerical attributes.

# Continuous Example

- Distinguish children from adults based on size
  - classes:  $\{a, c\}$ , attributes: height [cm], weight [kg]
  - training examples:  $\{h_i, w_i, y_i\}$ , 4 adults, 12 children
- Class probabilities:  $P(a) = \frac{4}{4+12} = 0.25$  ;  $P(c) = 0.75$
- Model for adults:
  - height  $\sim$  Gaussian with mean, variance
  - weight  $\sim$  Gaussian  $(\mu_{w,a}, \sigma_{w,a}^2)$
  - assume height and weight independent
- Model for children: same, using  $(\mu_{h,c}, \sigma_{h,c}^2), (\mu_{w,c}, \sigma_{w,c}^2)$

# Continue Example

$$P(a) = \frac{4}{4+12} = 0.25 ; P(c) = 0.75$$

$$p(h_x|c) = \frac{1}{\sqrt{2\pi}\sigma_{h,c}^2} \exp - \frac{1}{2} \left( \frac{(h_x - \mu_{h,c})^2}{\sigma_{h,c}^2} \right)$$

$$p(w_x|c) = \frac{1}{\sqrt{2\pi}\sigma_{w,c}^2} \exp - \frac{1}{2} \left( \frac{(w_x - \mu_{w,c})^2}{\sigma_{w,c}^2} \right)$$

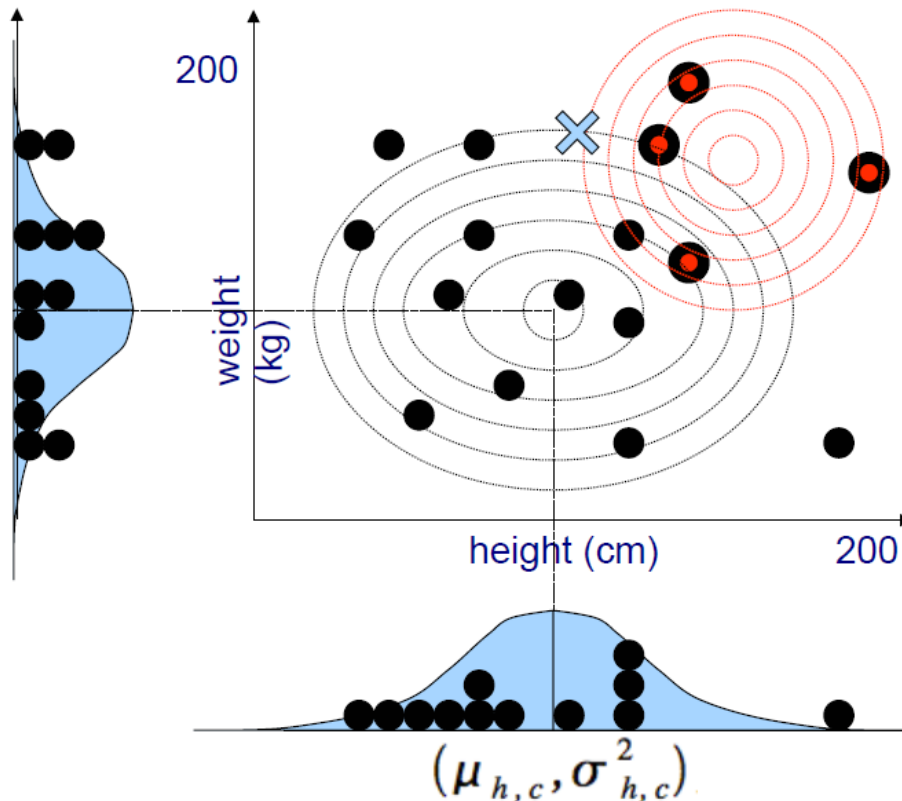
$$p(h_x|a) = \frac{1}{\sqrt{2\pi}\sigma_{h,a}^2} \exp - \frac{1}{2} \left( \frac{(h_x - \mu_{h,a})^2}{\sigma_{h,a}^2} \right)$$

$$p(w_x|a) = \frac{1}{\sqrt{2\pi}\sigma_{w,a}^2} \exp - \frac{1}{2} \left( \frac{(w_x - \mu_{w,a})^2}{\sigma_{w,a}^2} \right)$$

$$P(x|a) = p(h_x|a) p(w_x|a)$$

$$P(x|c) = p(h_x|c) p(w_x|c)$$

$$P(a|x) = \frac{P(x|a)P(a)}{P(x|a)P(a) + P(x|c)P(c)}$$



# For the Previous Example

[illegible]

# Assume Normal Distribution

- Let  $x_1, x_2, \dots, x_n$  be the values of a numerical attribute in the training data set.

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\sigma = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2$$

$$f(w) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(w-\mu)^2}{\sigma^2}}$$

# Apply Naïve Bayes Classifier

- For examples,

$$f(\text{temperature} = 66 \mid \text{Yes}) = \frac{1}{\sqrt{2\pi}(6.2)} e^{-\frac{(66-73)^2}{2(6.2)^2}} = 0.0340$$

- Likelihood of Yes =  $\frac{2}{9} \times 0.0340 \times 0.0221 \times \frac{3}{9} \times \frac{9}{14} = 0.000036$
- Likelihood of No =  $\frac{3}{5} \times 0.0291 \times 0.038 \times \frac{3}{5} \times \frac{5}{14} = 0.000136$
- $P(\text{Yes} \mid \mathbf{x}') < P(\text{No} \mid \mathbf{x}')$ , we label  $\mathbf{x}'$  to be “No”.



# Spam Filter Example

- Separate spam from valid email, attributes = words

D1: "send us your password" **spam**  
 D2: "send us your review" **ham**  
 D3: "review your password" **ham**  
 D4: "review us" **spam**  
 D5: "send your password" **spam**  
 D6: "send us your account" **spam**

new email: "review us now"

| P (spam) = 4/6   P (ham) = 2/6 |     |          |
|--------------------------------|-----|----------|
| spam                           | ham |          |
| 2/4                            | 1/2 | password |
| 1/4                            | 2/2 | review   |
| 3/4                            | 1/2 | send     |
| 3/4                            | 1/2 | us       |
| 3/4                            | 1/2 | your     |
| 1/4                            | 0/2 | account  |

$$P(\text{review us} | \text{spam}) = P(0, 1, 0, 1, 0, 0 | \text{spam}) = (1 - \frac{2}{4})(\frac{1}{4})(1 - \frac{3}{4})(\frac{3}{4})(1 - \frac{3}{4})(1 - \frac{1}{4})$$

$$P(\text{review us} | \text{ham}) = P(0, 1, 0, 1, 0, 0 | \text{ham}) = (1 - \frac{1}{2})(\frac{2}{2})(1 - \frac{1}{2})(\frac{1}{2})(1 - \frac{1}{2})(1 - \frac{0}{2})$$

$$P(\text{ham} | \text{review us}) = \frac{0.0625 \times 2/6}{0.0625 \times 2/6 + 0.0044 \times 4/6} = 0.87 \text{ (note identical example)}$$

# Problems with Naïve Bayes

- Zero-frequency problem

- any mail containing “account” is spam:  $P(\text{account}|\text{ham}) = 0/2$
- solution: never allow zero probabilities

- Laplace smoothing: add a small positive number to all counts:

$$P(w|c) = \frac{\text{num}(w, c) + \epsilon}{\text{num}(c) + 2\epsilon}$$

- may use global statistics in place of  $\epsilon$ :  $\text{num}(w) / \text{num}$
- very common problem (Zipf's law: 50% words occur once)

- Assumes word independence

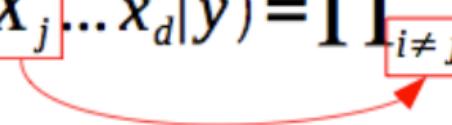
- every word contributes independently to  $P(\text{spam}|\text{email})$
- fooling NB: add lots of “hammy” words into spam email

# Dealing with Missing Data

- Suppose don't have value for some attribute  $X_j$ 
  - applicant's credit history unknown
  - some medical test not performed on patient
  - how to compute  $P(X_1=x_1 \dots X_j=? \dots X_d=x_d | y)$

- Easy with Naïve Bayes

- ignore attribute in instance where its value is missing

$$P(x_1 \dots \boxed{X_j} \dots x_d | y) = \prod_{\boxed{i \neq j}}^d P(x_i | y)$$


- compute likelihood based on observed attributes
  - no need to “fill in” or explicitly model missing values
  - based on conditional independence between attributes

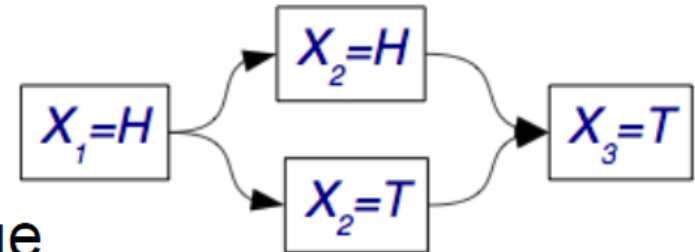
# Missing Data Example

- Ex: three coin tosses: Event =  $\{X_1=H, X_2=?, X_3=T\}$

- event = head, unknown (either head or tail), tail

- event =  $\{H, H, T\} + \{H, T, T\}$

- $P(\text{event}) = P(H, H, T) + P(H, T, T)$



- General case:  $X_j$  has missing value

$$P(x_1 \dots \boxed{x_j} \dots x_d | y) = P(x_1 | y) \cdots \boxed{P(x_j | y)} \cdots P(x_d | y)$$

$$\sum_{\boxed{x_j}} P(x_1 \dots \boxed{x_j} \dots x_d | y) = \sum_{x_j} P(x_1 | y) \cdots \boxed{P(x_j | y)} \cdots P(x_d | y)$$

$$= P(x_1 | y) \cdots \boxed{\left[ \sum_{x_j} P(x_j | y) \right]} \cdots P(x_d | y)$$

$$= P(x_1 | y) \cdots \boxed{1} \cdots P(x_d | y)$$

# Conclusions

- Naïve Bayes based on the independence assumption
  - Training is very easy and fast; just requiring considering each attribute in each class separately
  - Test is straightforward; just looking up tables or calculating conditional probabilities with normal distributions
- A popular generative model
  - Performance competitive to most of state-of-the-art classifiers even in presence of violating independence assumption
  - Many successful applications, e.g., spam mail filtering
  - A good candidate of a base learner in ensemble learning
  - Apart from classification, naïve Bayes can do more...